



Physiology Of Hormones

Definitions and Functions

Dr. Nevine Jasim Karash

MBChB., MSc. Medical Physiology

Department of Medical Physiology and Microbiology

College of Medicine

Hawler Medical University HMU

2025-2026

nevine.jasim@hmu.edu.krd

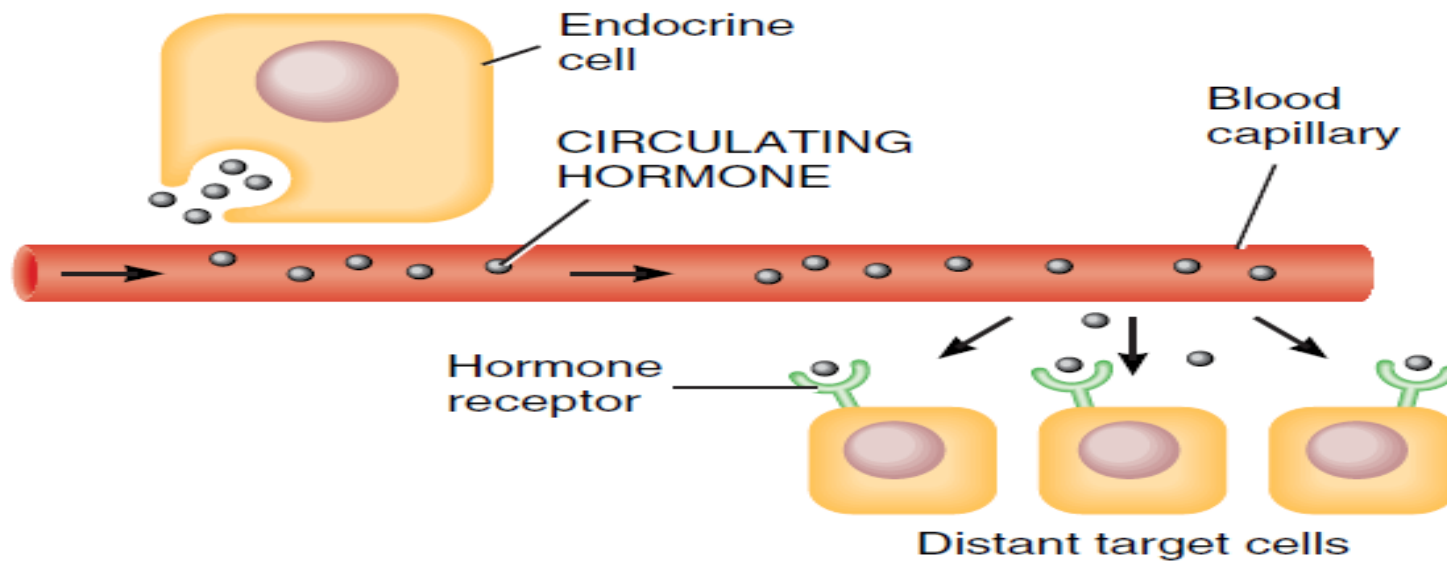
Learning Outcomes:

- Define a hormone.
- Differentiate between circulation and local hormones.
- Classify hormones according to their solubility.
- Explain the mechanism of action hormones.

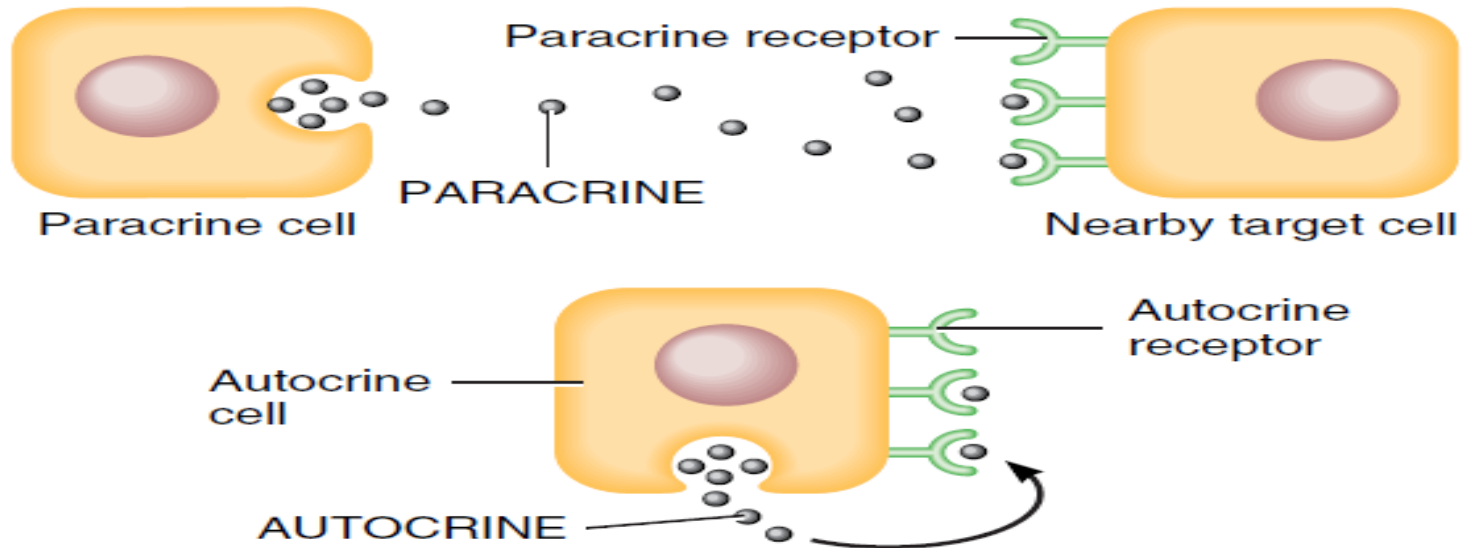
- A **hormone** is a mediator molecule that is released in one part of the body but regulates the activity of cells in other parts of the body.
- Most hormones enter interstitial fluid and then the bloodstream.
- The circulating blood delivers hormones to cells throughout the body.
- Both neurotransmitters and hormones exert their effects by binding to receptors on or in their “target” cells.

Circulating and Local Hormones:

- Most endocrine hormones are **circulating hormones** which means they pass from the secretory cells that make them into interstitial fluid and then into the blood.
- Other hormones, termed **local hormones**, act locally on neighboring cells or on the same cell that secreted them.
 - Local hormones acting on neighboring cells are called  paracrine.
 - Local hormones acting on the same cell that secreted them  autocrine.



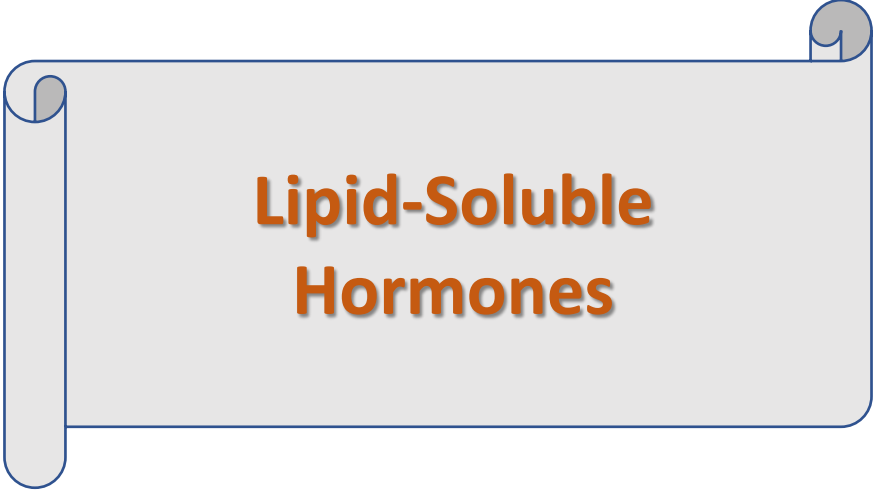
(a) Circulating hormones



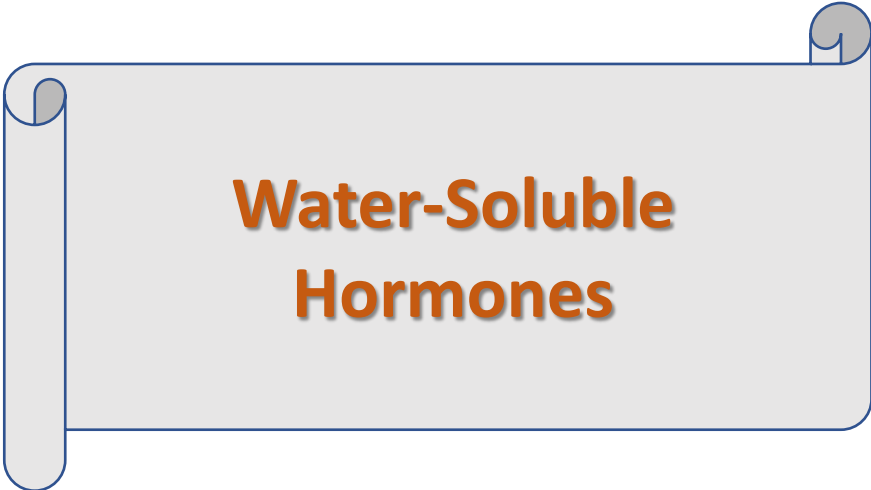
(b) Local hormones (paracrines and autocrines)

Chemical Classes of Hormones

Chemically, hormones can be divided into two broad classes:

A light gray scroll with a blue border, partially unrolled from the top right and bottom left corners.

**Lipid-Soluble
Hormones**

A light gray scroll with a blue border, partially unrolled from the top right and bottom left corners.

**Water-Soluble
Hormones**

Lipid-Soluble Hormones

- 1) **Steroid hormones** are derived from cholesterol.
- 2) **Two thyroid hormones (T3 and T4)** are synthesized by attaching iodine to the amino acid tyrosine.
- 3) **The gas nitric oxide (NO)** is both a hormone and a neurotransmitter.

Water-Soluble Hormones

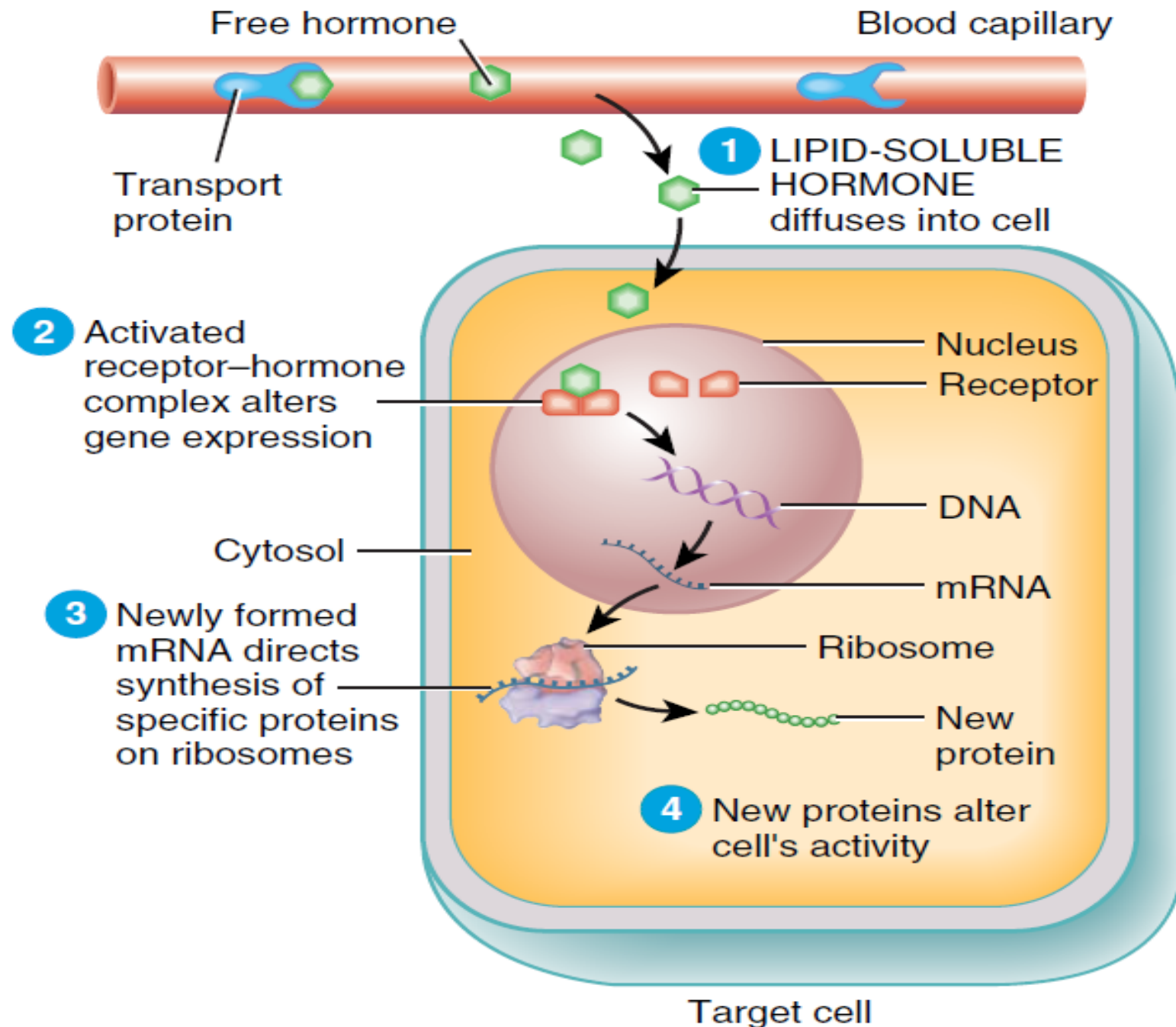
- 1) **Amine hormones:** they retain an amino group ($-\text{NH}_3$). EX: epinephrine, norepinephrine, histamine, Serotonin.
- 2) **Peptide hormones and protein hormones:** peptide hormones as antidiuretic hormone ADH and oxytocin; protein hormones like insulin.
- 3) **The eicosanoid hormones:** The two major types of eicosanoids are prostaglandins (PGs) and leukotrienes (LTs).

Hormone Transport in the Blood

- Most water-soluble hormone molecules circulate in the watery blood plasma in a “free” form (not attached to other molecules).
- Most lipid-soluble hormone molecules are bound to transport proteins.
- The transport proteins, which are synthesized by cells in the liver, They make lipid-soluble hormones temporarily water-soluble.

Action of Lipid-Soluble Hormones

1. A free lipid-soluble hormone molecule **diffuses** from the blood then through the lipid bilayer into a cell.
2. The hormone **binds to receptor** within the cytosol or nucleus, receptor-hormone complex **alters gene expression**: It turns specific genes of the nuclear DNA on or off.
3. As the DNA is transcribed, **new mRNA forms**, leaves the nucleus, and enters the cytosol.
4. There, it directs **synthesis of a new protein**.
5. The new proteins alter the cell's activity and cause the responses typical of that hormone.



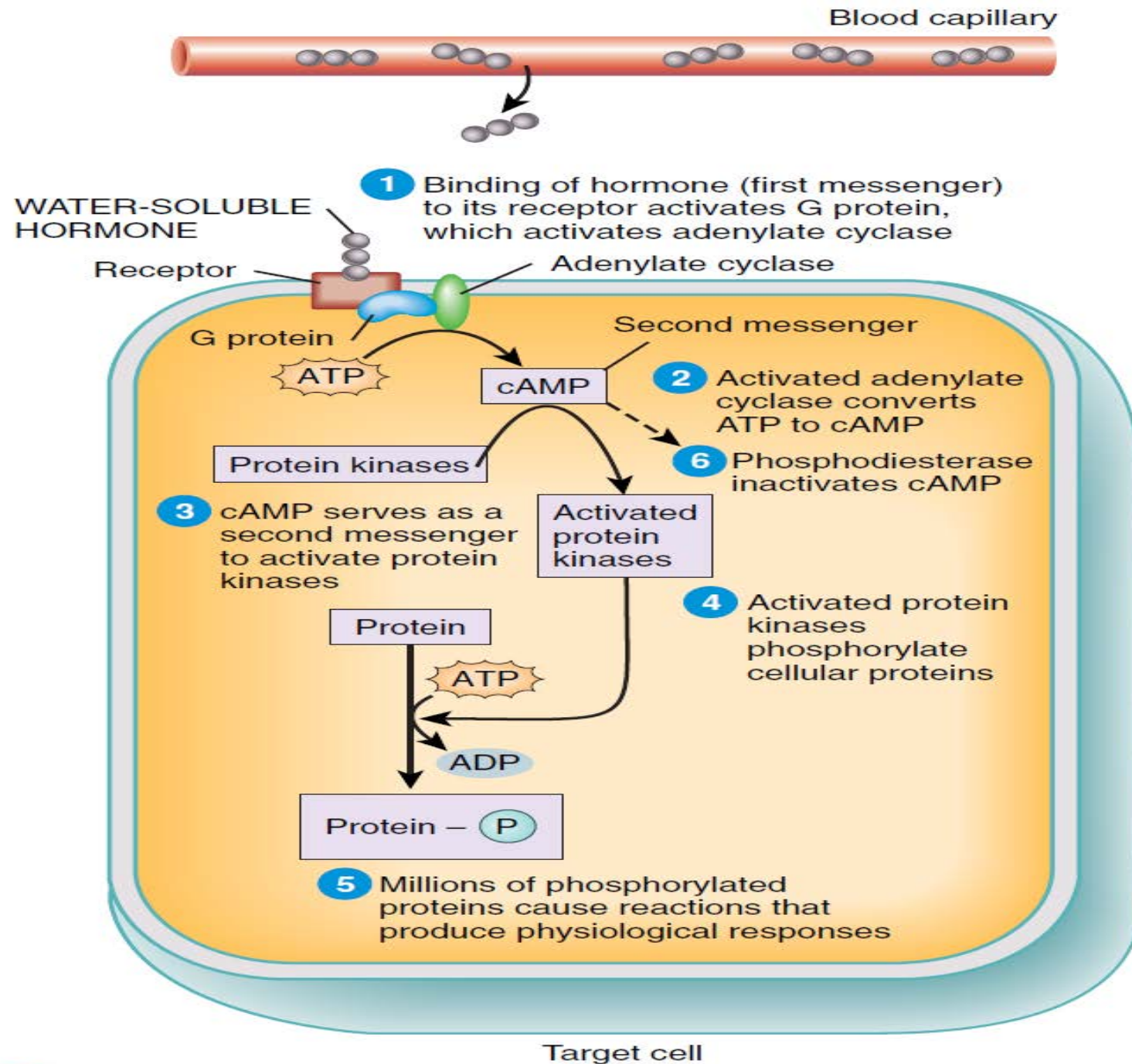
Action of Water-Soluble Hormones

- 1) A water-soluble hormone (the first messenger) diffuses from the blood then binds to its receptor at the exterior surface of a target cell. The hormone–receptor complex activates a membrane protein called a G protein. The activated G protein in turn activates adenylate cyclase.
- 2) Adenylate cyclase converts ATP into cyclic AMP (cAMP).
- 3) Cyclic AMP (the second messenger) activates protein kinases. A protein kinase is an enzyme that phosphorylates (adds a phosphate group to) other cellular proteins (such as enzymes). The donor of the phosphate group is ATP, which is converted to ADP

4) Phosphorylation activates some of proteins and inactivates others.

5) Phosphorylated proteins in turn cause reactions that produce physiological responses: Ex.

- triggering glycogen synthesis
- breakdown of triglyceride
- inactivate an enzyme needed for glycogen synthesis.



Questions and
Comments?